

PH 702: Introduction to Biostatistics, Fall 2026



Credits: 3

Meeting times: (Section 401) Mon 5:30–7:20 pm @Zilber Triplex,
(Section 802) Thur 2:30–4:20 pm @Zilber 240,
(Section 801) Thur 7:30–9:20 pm @Zilber 240, OR
(Sections 202 and 901) asynchronous.

Instructor

Instructor name: Zhiyang Zhou

Instructor email: zhou67@uwm.edu

Office hours: (Section 401) Mon 4:00–5:00 pm @Zilber 376

Teaching Assistant

TA name: Don Cramer

TA email: dkramer@uwm.edu

Office hours: by appointment

Welcome Statement

Welcome to PH 702! This course introduces the fundamental concepts and methods of biostatistics and their applications in public health and health research. Through lectures, hands-on exercises, and real-world examples, you will develop practical skills in analyzing, interpreting, and communicating quantitative information. I look forward to working with you and supporting your learning throughout the semester.

Course Overview

This course introduces students to applied statistics with basic definitions and many examples. Topics will include Types of Data, Data Visualization, Descriptive Statistics, Probability, Sampling, and Statistical Inference, which will include confidence intervals and hypothesis testing. Many of the calculations and graphical representations will be done with SAS.

Course Objectives

- Provide students with a foundational understanding of biostatistical concepts and their roles in public health research and practice.
- Introduce principles of data description, visualization, probability, sampling, and statistical inference.
- Develop students' understanding of commonly used statistical methods and the assumptions underlying their appropriate application.
- Provide practical experience in selecting and applying statistical methods to public health and health-related data.
- Develop foundational skills in using statistical software to manage, analyze, and visualize data.

- Strengthen students' ability to interpret and communicate statistical findings in the context of public health research.

Student Learning Outcomes

By the end of the course, students will be able to:

- Explain the role of biostatistics and quantitative evidence in public health research and practice.
- Identify and distinguish types of data, measurement scales, and common study and sampling approaches.
- Summarize and visualize public health data using appropriate descriptive statistics and graphical methods.
- Apply fundamental concepts of probability, random variation, sampling distributions, and commonly used probability distributions.
- Formulate research questions and translate them into appropriate statistical hypotheses.
- Select and apply appropriate statistical methods, including confidence intervals, hypothesis tests, methods for categorical data, regression, and analysis of variance.
- Assess key assumptions underlying statistical methods and identify appropriate alternatives when assumptions are not adequately met.
- Use SAS to manage, analyze, and visualize public health and health-related data.
- Interpret statistical results in the context of substantive public health questions, including measures of uncertainty and statistical significance.
- Communicate statistical methods, results, and conclusions effectively through written reports, tables, figures, and oral presentations.

Assessment

Students will be evaluated through regular assignments and exams. Assessment is aligned with the stated learning outcomes and emphasizes reproducibility, practical proficiency, and clear data communication.

Prerequisites and/or Special Skills Required

Graduate student status; Math 116 (P) with B or better, or equivalent, or consent of instructor.

Course Modality and Format

- Format: lectures.
- Modality: onsite (section 401)/online and asynchronous (section 202).
- Detail: Active participation during class time is essential, as learning will take place through live instruction, hands-on coding, group discussions, and collaborative problem-solving.
 - Interaction with the instructor: The instructor will lead live lectures and demonstrations during each class session, introducing new concepts and walking through SAS code examples in real time. Students are expected to ask questions, share screens when needed, and engage in discussion during class. Office hours (by appointment) will be held on lecture days for one-on-one support or deeper technical questions.

- Interaction with peers: Students will work in breakout groups, pairs, or small teams during class to complete coding challenges, troubleshoot data issues, and review visualizations. Respectful, professional interaction is expected at all times, whether in discussion or screen-sharing activities.
- Interaction with course content: Lecture notes will be posted before each lecture to support self-directed learning. Students are expected to review these materials in advance so that live sessions can focus on hands-on coding, clarification, and collaborative exercises. Course content is reinforced through regular assignments and in-class application, with independent practice encouraged to build proficiency in SAS.

Course Materials

- Required readings: Materials regularly posted at [the instructor's homepage](#) and Canvas.
- Recommended Readings:
 - D. Moore, G. McCabe, & B. Craig. (2017). *Introduction to the Practice of Statistics*, 9th Ed. New York: W. H. Freeman & Co.
 - L. Delwiche & S. Slaughter. (2012). *The Little SAS Book: A Primer*, 5th Ed. Cary, NC: SAS Institute Inc.

Time Investment

This is 3-credit course, which requires a minimum of 144 hours of total time commitment over the semester. This translates to approximately 9–10 hours of work per week across 14 weeks, including both in-class and out-of-class activities. Each week, students should expect to spend their time roughly as follows: 2.5 hours in lectures, 2–3 hours reviewing pre-posted lecture materials, reading documentation, or exploring example code, and 4.5 hours completing assignments.

Technology Requirements

Each student is expected to have regular access to a computer with a reliable internet connection. This course makes extensive use of SAS.

Assignments & Grading

Assignment and Grading Policies

- Homework (60%): Homework will be assigned at the end of each lecture and will be due at the beginning of the following lecture. Assignments should be submitted in a clear and organized fashion through Canvas.
- Exams (40%): The midterm exam is scheduled for Monday, Oct. 26th during class time. The schedule for the final exam is TBD.

Grading Scheme

Assignment/Quantity	Point Value
Homework (12; 5 points each)	60

Assignment/Quantity	Point Value
Midterm (1)	20
Final (1)	20
TOTAL	100

Grading Scale

Grade	Scale
A	93–100
A-	90–92
B+	87–89
B	83–86
B-	80–82
C+	77–79
C	73–76
C-	70–72
D+	67–69
D	63–66
D-	60–62
F	0–59

Policies

Course Policies

- Attendance Policies (for Section 401): Though there is no penalty on absence, it is better to be present in the designated session punctually. Since the course will be delivered without any forms of recording, there is no alternative way of attendance.
- Makeup or Late work. No late submissions will be accepted. Timely submissions will be graded and returned within one week. Students with personal or medical emergencies should contact their instructor as soon as possible.
- Academic Integrity Policy. Academic integrity involves taking responsibility for one's own work, demonstrating honesty in academic efforts, and showing respect for the intellectual contributions of others. As members of the university community, all students are expected to understand and uphold these principles. The Joseph J. Zilber College of Public Health remains committed to maintaining high standards of academic integrity. Each student is expected to contribute to this commitment by avoiding academic misconduct, including but not limited to impersonation, plagiarism, and the use of unauthorized materials. In situations where expectations are unclear (such as whether collaboration, tools, or specific resources are permitted), seeking clarification from the instructor is encouraged. Asking questions reflects a commitment to integrity and a desire to engage responsibly with the academic process. Useful resources can be found at UWM [information on academic misconduct](#). Also, please refer to [CETL's tips on academic integrity in online learning for instructors and students](#).

- Policy on Artificial Intelligence. Students are permitted to use Generative Artificial Intelligence (GenAI) tools, such as [Microsoft Copilot](#), to support their learning. These tools may assist with understanding course content, brainstorming ideas, and revising written drafts. However, final submissions for assignments and projects must reflect students' own work and should not be entirely generated by GenAI. If GenAI tools are used, proper citation is required. For guidance on citing AI-generated content, please refer to the resource on [using and citing GenAI](#), provided by UWM Libraries. Taking credit for any work that is not original, whether in whole or in part, constitutes academic misconduct and may lead to disciplinary action under [UWS Chapter 14](#). While GenAI can be helpful, it is essential to use them responsibly and ethically. AI-generated content may be biased, inaccurate, incomplete, or otherwise unreliable. Students must critically evaluate any output from GenAI before incorporating it into academic work.

University Policies

- Statement on Hate/Bias Incidents: UWM is committed to creating and supporting a campus climate that is respectful and supportive of all who study, live, or work on campus, or participate in campus activities. UWM students and employees who have been subjected to, or who have witnessed, an incident of bias or hate are encouraged to submit a [hate/bias incident report](#). UWM can provide support or resources to those involved in the incident. Find more information about hate- or bias-motivated incidents, as well as UWM's Discriminatory Conduct Policy and other resources, on UWM's [hate/bias webpage](#).
- UWM Syllabus link: Include a link to the [Syllabus Links](#) page, which details policies pertaining to students with disabilities, absences due to religious observation, students called to active military duty, incompletes, discriminatory conduct, Title IX, academic misconduct, complaint procedures, grade appeal procedures, LGBT+ resources, and final exam policies.

Resources

Academic Supports

- [Accessibility Resource Center](#). Dedicated to providing equal access for students with disabilities to UWM's academic, social, cultural and recreational programs.
- Canvas Support: Visit [UWM Canvas Student Support](#) or reference the official [Canvas Student Guide](#).
- [Student Success Center](#): Explore tutoring options for your courses or attend a supplemental instruction session. Student Success Center also offers one-on-one academic skills tutoring sessions to help students with time management and strategies for notetaking, studying, and test taking.
- [Textbooks via Panther Shop](#). Find out what materials you need for your courses.
- [UWM Libraries](#). Conduct your own research using a vast array of databases, seek help from librarians, reserve rooms for group projects, find a quiet study space, stop by the coffee shop.
- [Writing Center](#). Meet confidentially, one-on-one, face-to-face or online in real time with writing specialists. Any subject, any project, any level.

Non-Academic Supports

[Support U](#). Any student in need, or students that face challenges that are barriers to their education, are encouraged to contact the Dean of Students (dos@uwm.edu) for support. Support U offers wrap-around holistic support for students, including basic needs, accessing the food pantry, emergency funding, case management, and connecting to resources, etc.

- [Emergencies](#): Includes crisis and after care, details on medical withdrawals, and UWM Police.
- [Financial Resources](#): Links to Student Financial Services, scholarship information, emergency grants, and the food pantry.
- [Health Resources](#): Medical and counseling services, community resource finder, the UWM Psychology clinic, survivor support and victim advocacy.
- [Guidance and Consultation](#). Details on appeals, parking, accommodations, and mechanisms for filing grievances.

Technology Resources

- [Student Technology Resources](#)

Careers

- [Student Experience and Talent](#)

Centers for Advocacy & Student Engagement (CASE)

- [First-Generation+ Resource Center](#)
- [LGBTQ+ Resource Center](#)
- [Military and Veterans Resource Center](#)
- [Women's Resource Center](#)

Multicultural Student Centers (MSCs)

- [American Indian Student Center](#)
- [Black Student Cultural Center](#)
- [Roberto Hernández Center](#)
- [Southeast Asian American Student Center](#)

Student Involvement

- [Event Calendar](#)
- [Student Association \(Student Government\)](#)
- [Student Involvement/Organizations](#)
- [Office of Undergraduate Research](#)

Important Dates

Event

First day of classes

Labor Day

Add deadline

Last day to drop without a "W"

Date

Sept. 2, 2026

Sept. 7, 2026

Sept. 9, 2026

Sept. 29, 2026

Event

Drop deadline
 No Class (Thanksgiving)
 Last day of classes
 Study Day
 Final exams begin
 Grade Deadline

Date

Nov. 8, 2026
 Nov. 25–29, 2026
 Dec. 14, 2026
 Dec. 15, 2026
 Dec. 16, 2026
 Jan. 4, 2027

Instructor statement of support

I know this course may look challenging at first glance. Learning to work with data takes time, practice, and persistence. Please don't hesitate to reach out if you're struggling, stuck, or just need a nudge in the right direction. I believe each of you can succeed in this class, and I'm committed to helping you get there. Let's work together to make this a productive and supportive semester.

Calendar

Date	Topic	Assignments
Sept. 14	Introduction, univariate data with plots, descriptive statistics	HW1
Sept. 21	Bivariate relationships, two- way tables	HW2
Sept. 28	Simple linear regression	HW3
Oct. 5	Producing data, sampling	HW4
Oct. 12	Sampling distributions	HW5
Oct. 19	Introduction to inference: confidence and significance	HW6
Oct. 26	Midterm exam	
Nov. 2	Inferences for means	HW7
Nov. 9	Inferences for proportions	HW8
Nov. 16	Inference for contingency tables	HW9
Nov. 23	Inference in simple linear regression	HW10
Nov. 30	Multiple regression	HW11
Dec. 7	ANOVA	HW12
Dec. 14	Recap	
TBD	Final exam	